Understanding CRISP-DM using Video Game Sales Data

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In this post, we'll get a deep understanding CRISP-DM with a video game Sales Data and give the answer of four analytical questions by using python:

- What revenue game sales generate and how many games releases every year?
- which publishers are leader by game revenue and games releases?
- Which game genre has highest releases and love by people?
- Can we make a regression model to predict the sales price?

Image source: Udacity

Let's understand CRISP-DM

There are a lot of different tools and techniques used for a large number of industry problems which falls under the domain of data science, However, there is a common process used to find any solutions in Data Science and this process is known as Cross-Industry Standard Process of Data Mining. This Process is an industry standard for analyzing data for years and it goes with six phases:

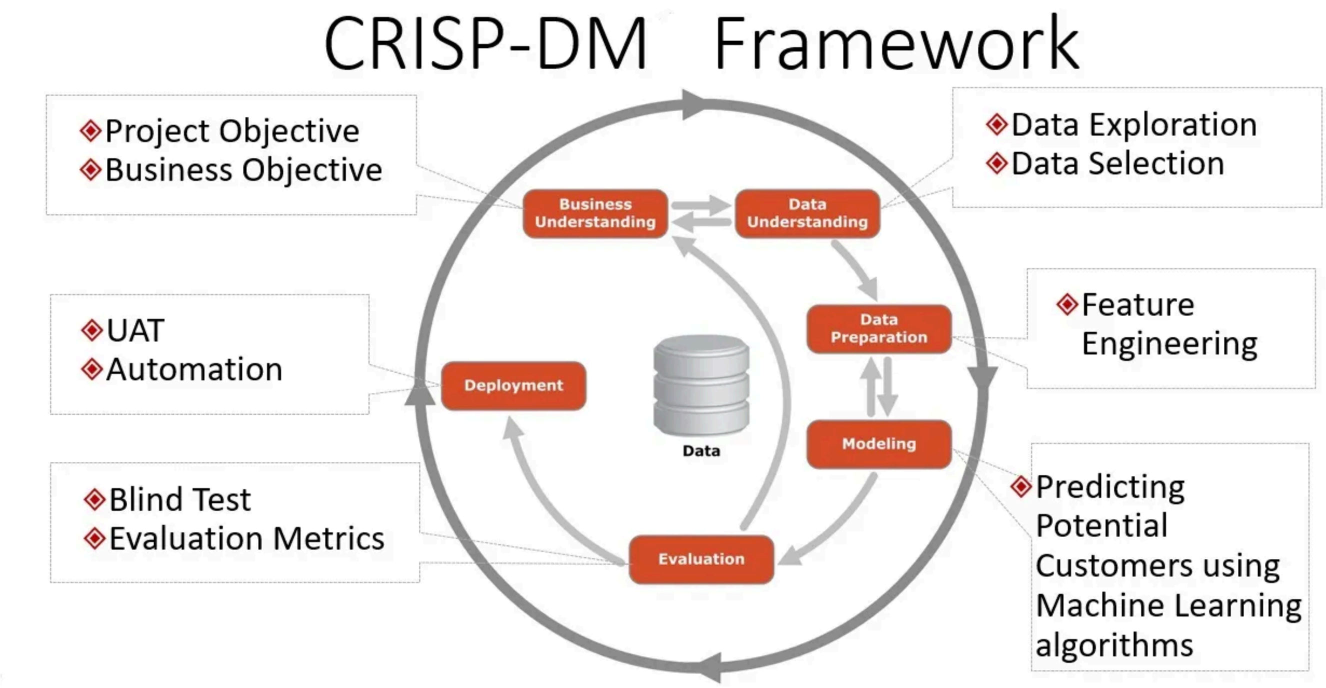


Image source: Google Image Search

Business Understanding

The first phase of CRISP-DM process is business understanding and this is all about knowing your business problems. You could be an e-commerce company which needs to optimize it deliver cost or give better product recommendations or you could be gaming company which is working to make more engaging games and so It is very important to ask the right question, for example:

How to get new customers?

Is the new composition of a drug is better than earlier?

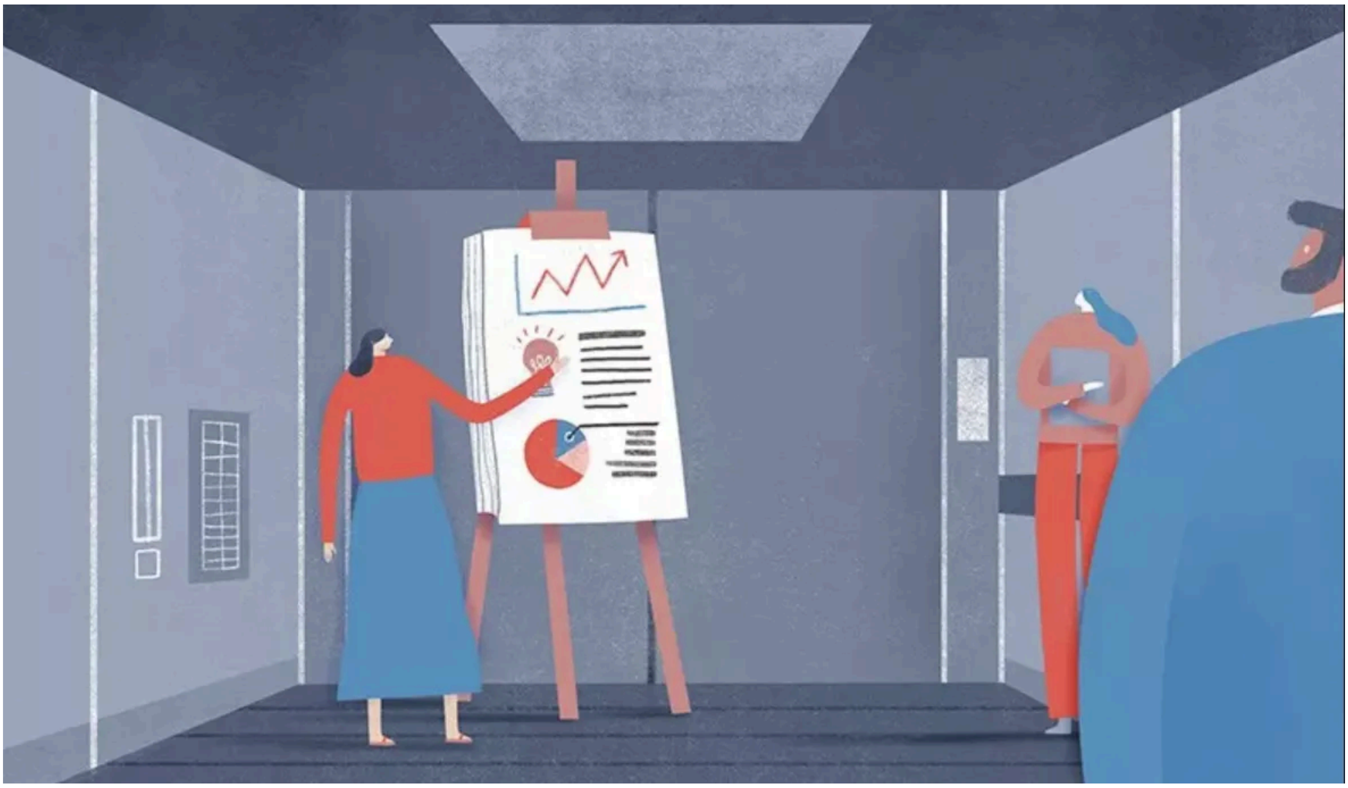


Image source: kraywiki.com

Data Understanding

The second phase of the CRISP-DM process is data understanding. Business these days generate a huge amount of data So understanding what data is required to solve our business problem is quite important.

Sometimes you have mountains of data and you need to dig through it and other time you may need to collect data, this means you need to understand what kind of data can give you insights?

This is very difficult to know ahead of time So business collects all possible data first so that they can later identify which data they can use to find their insights.

Data Understanding and Preparation

This dataset which we are using contains a list of video games with sales greater than 100,000 copies. It scraped from vgchartz.com and available on kaggle.

	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
0	1	Wii Sports	Wii	2006.0	Sports	Nintendo	41.49	29.02	3.77	8.46	82.74

Image source: Udacity

By looking at the first row, we can verify our dataset as below:

- Rank — Ranking of overall sales
- Name — The games name
- Platform — Platform of the game's release (i.e. PC,PS4, etc.)
- Year — Year of the game's release
- Genre — Genre of the game
- Publisher — Publisher of the game
- NA_Sales — Sales in North America (in millions)
- EU_Sales — Sales in Europe (in millions)
- JP_Sales — Sales in Japan (in millions)

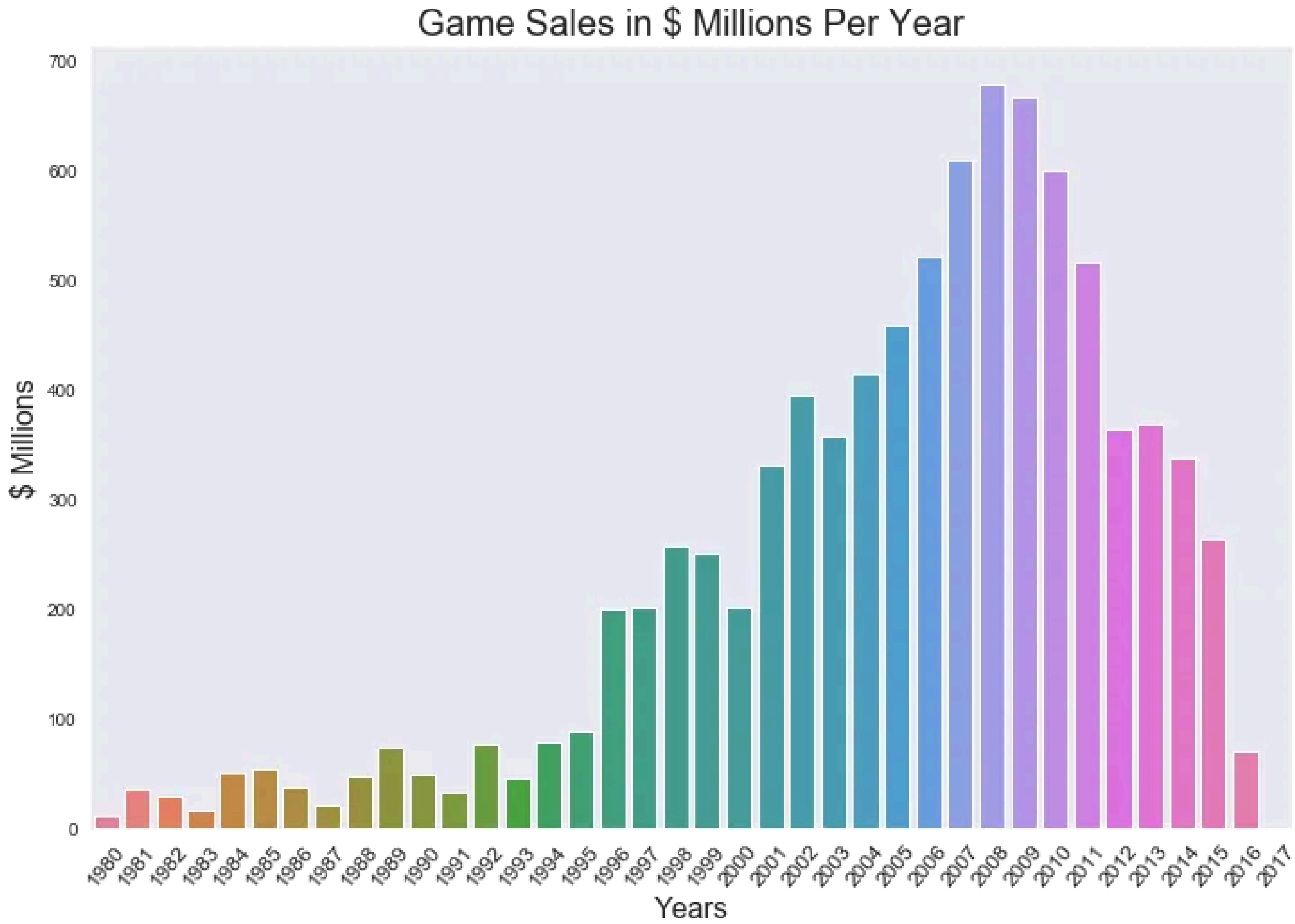
- Other_Sales — Sales in the rest of the world (in millions)
- Global_Sales — Total worldwide sales.

Below we check the number of games (rows) and the number of unique publishers, platforms, and genres to get an idea of how our the games in the dataset are distributed categorically

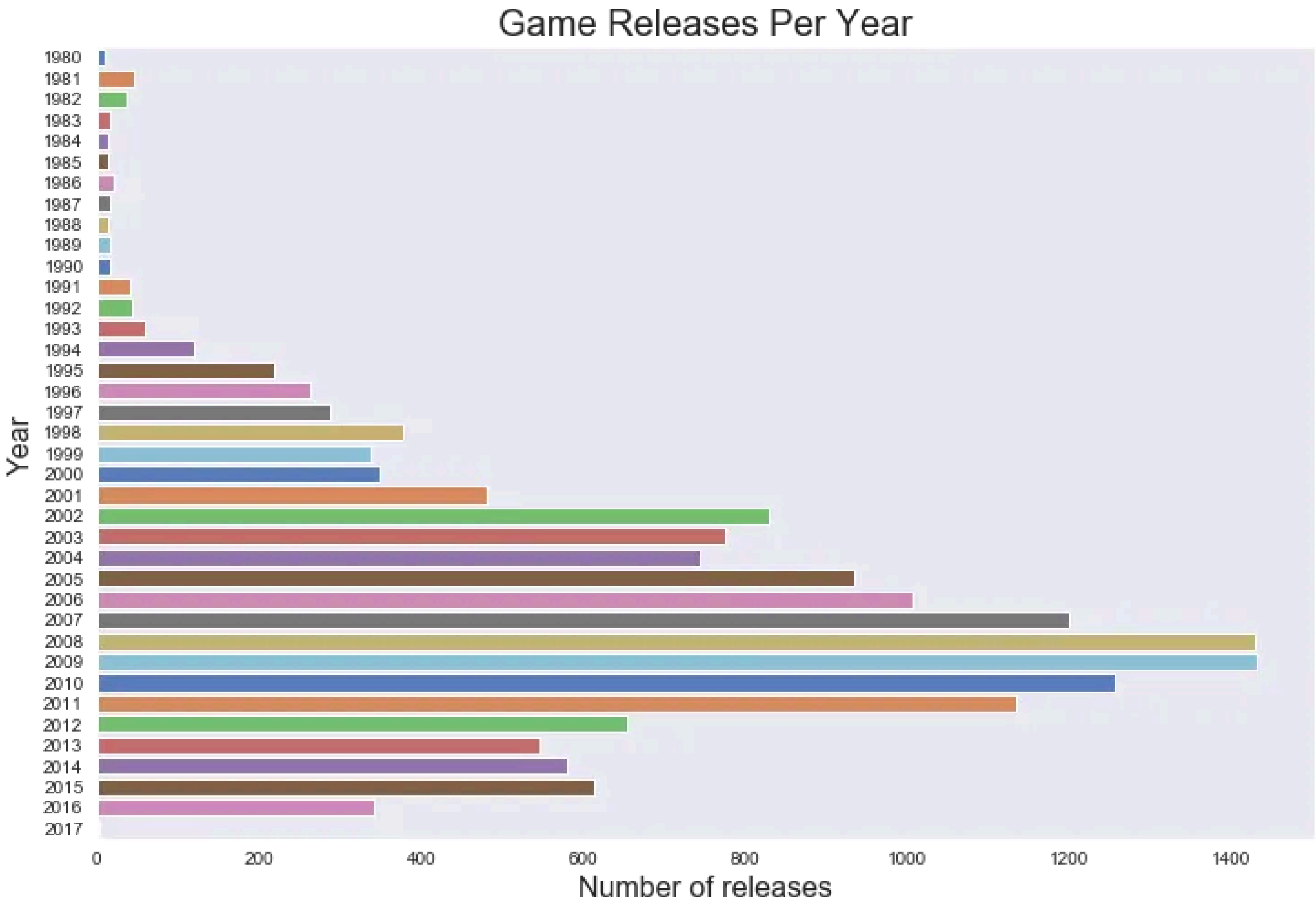
- Number of games: 16598
- Number of publishers: 579
- Number of platforms: 31
- Number of genres: 12

Now, as we are in data understanding and preparation step, we'll answer all our questions mentioned in above chronological order as this is the part of EDA.

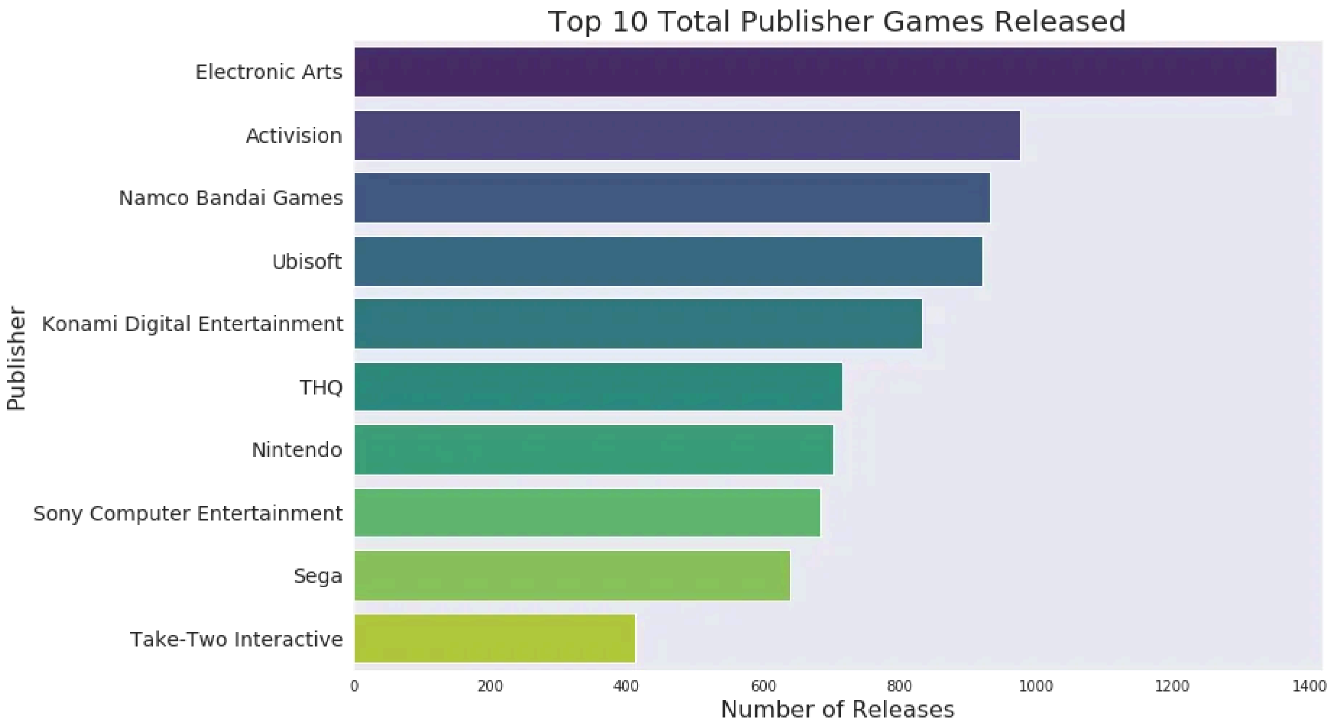
So, Here is the revenue of games sales by year and we can see clearly that 2008 is the year when games sales was on top.

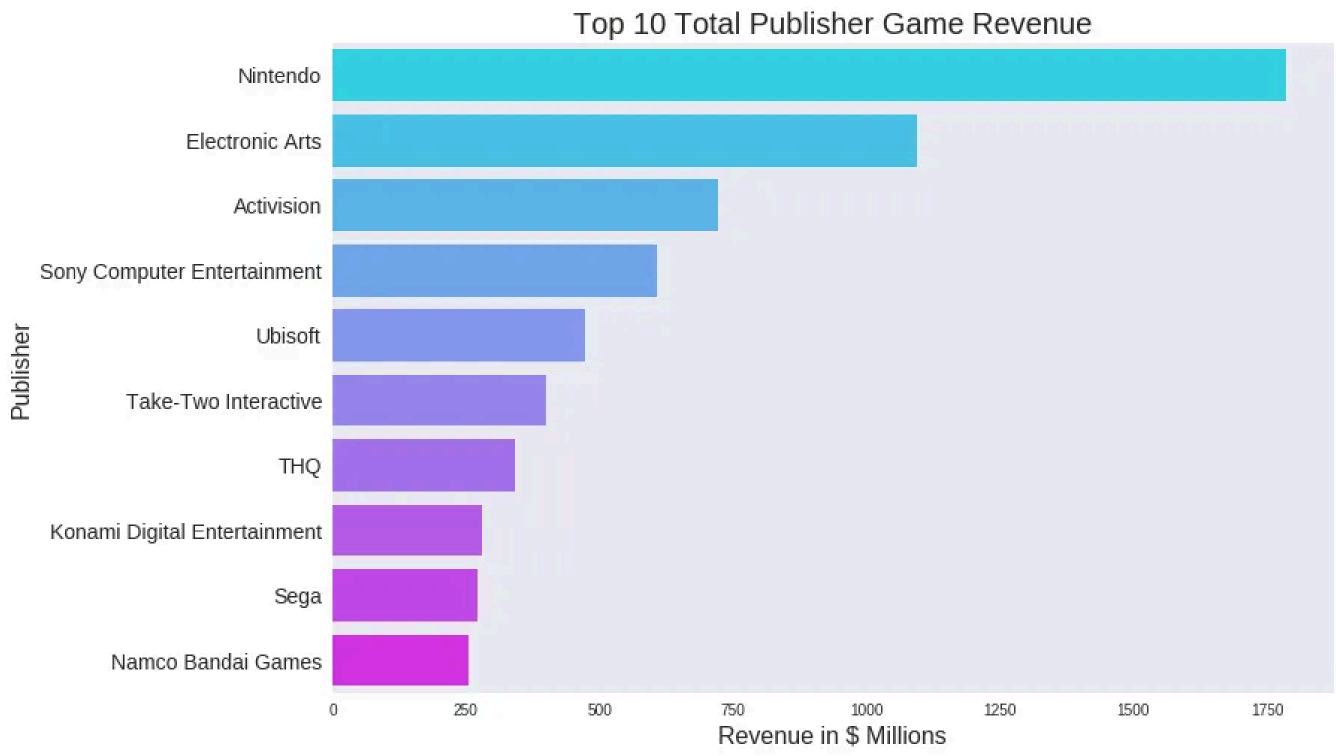


Our second part of first question was for number of game releases. we see by below bar chart that video games releases gone down in last few years and 2008 and 2009 are peak years.



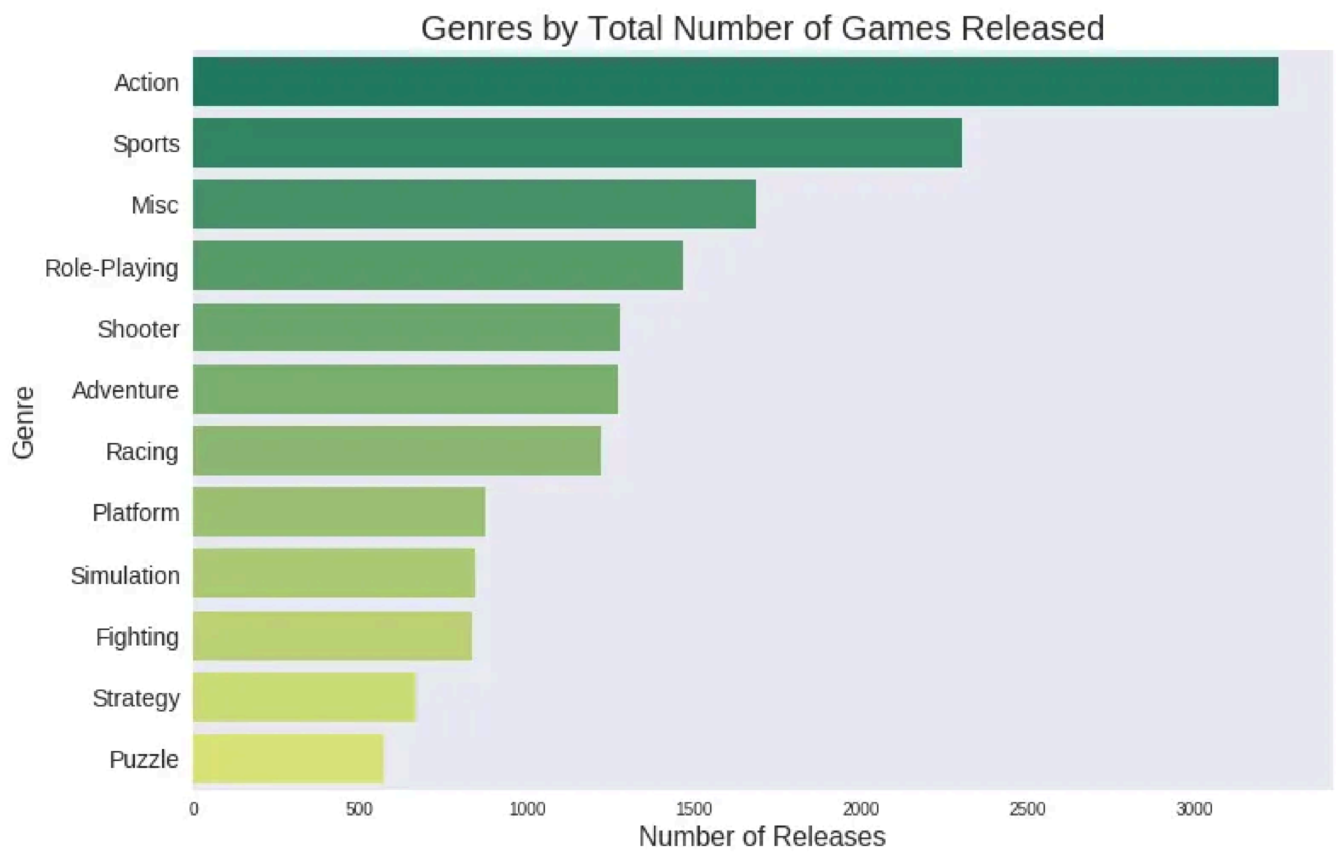
We are also interested in know which publisher is market leader in terms of revenue and releases.

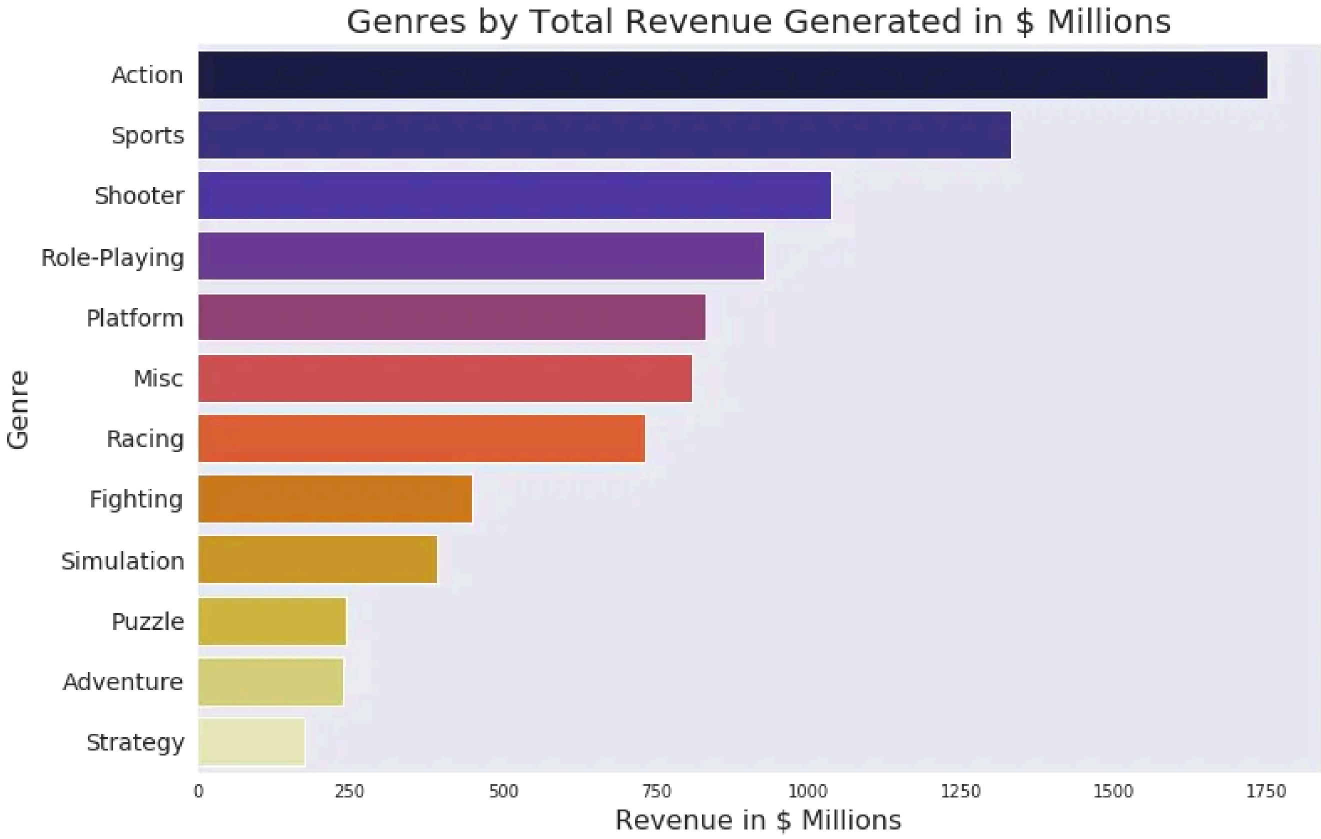




Now we have top 10 publisher by revenue and releases. Electronic Arts and Nintendo are leading by releases and revenue receptively.

Our third question was most loved genre and that is action. we are saying this on the bases of games releases and revenue but I also see the current trend where PUBG and Fortnite are two popular action games.





Modeling

Once we are done with data preparation process like filling the NA values, feature engineering and feature selection, we are ready to model our data as a regression or classification task based or target label. In our cases we have to predict sales value so this is regression task. we've build linear regression, xgboost and ensemble model to predict the sales.

Evaluation

Evaluation is the 2nd last step of CRISP-DM process when we check the accuracy of our model and go to previous stages of feature engineering or feature selection. Our console price prediction model Training score: 0.9999716232951564 and Testing score: 0.999958555291659 which seems like very good.

Here is the comparison of actual value and predicted values by our model.

	LINEAR_REGRASSOR	XGBOOST	ENSEMBLE	ACTUAL_PRICE
9317	0.130333	0.133437	0.133394	0.14
14835	0.020339	0.028703	0.028657	0.03
9752	0.120335	0.119160	0.119116	0.12
10251	0.110324	0.114278	0.114234	0.11
16565	0.010337	0.011232	0.011186	0.01
2821	0.720267	0.720812	0.720781	0.72
5756	0.310330	0.309910	0.309870	0.31
7303	0.220281	0.214665	0.214623	0.22
6964	0.230295	0.241061	0.241019	0.23
6821	0.240330	0.243782	0.243741	0.24

Deployment

Finally, when we are done with all our evaluation and testing, we can deploy the model on production to use.

In this article, we approach potential analytical questions for video game industry by CRISP-DM model on video games console sales datasets.

The source code with the corresponding Jupyter Notebook for this article is posted at my Github

Data Science · Crisp Dm · Linear Regression · Video Game Analysis

Written by Bharat Mishra

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